

01 / EXECUTIVE DECISION

Image access works. Clinical trust is unproven.

A completed comparison of GPT imaging workflows and pinned MedGemma, using existing doctors' reports as the reference. Research evidence; no live clinician adjudication.

970

Frozen assignments closed

919

Qualified outcomes

120

Main imaging studies

Recommendation: retain research engineering; hold clinical deployment

Successful image delivery and structured answers establish technical feasibility. Poor report-positive finding resolution, input limitations and small reference counts do not support autonomous patient-care use or radiologist equivalence.

DECISIONS SUPPORTED BY THE EVIDENCE

Decision	Basis
Continue a bounded research workflow	All three GPTs captured qualified direct responses on 120/120 main studies. Identity and input audits remain essential.
Hold the tested diagnostic workflows	Pneumothorax resolution: 0-1 of 10 positives. CT nodule resolution: 0-3 of 19 positives.
Do not add a specialist by default	MedGemma consultation has no established general benefit versus self-review, with additional diagnostic components.
Investigate a narrow MRI endpoint	White-matter assertion concordance is promising in a tiny, source-conflicted sample; fresh validation is absent.

[E1-E4] Qualified outcomes measure execution, not diagnostic accuracy. Results concern these tested inputs and budgets.

02 / STUDY COVERAGE

A completed study, with distinct evidence tiers.

Repeated runs and multiple workflow assignments are not additional independent patients. Main, development and fresh cohorts remain separate.

Evidence tier	Study count	What it supports
Main cohort	120	60 chest radiographs (CXR), 40 chest CT volumes, 20 MRI studies.
Fresh cohort	30	10 per modality; Sol and MedGemma only. Not all models or role workflows.
Development	20	Feasibility and setup checks; excluded from primary performance.

QUALIFIED DIRECT-INPUT RESPONSES

Model	CXR / 60	CT / 40	MRI / 20
gpt-6-astra	60/60	40/40	20/20
gpt-6-luna	60/60	40/40	20/20
gpt-6.1-sol	60/60	40/40	20/20
MedGemma 1.5 4B NF4	60/60	32/40	17/20

Completion is a prerequisite, not a diagnostic verdict

The frozen engineering gate requires at least 95% qualified responses, a Wilson 95% lower bound of at least 80%, and clean identity/isolation. GPT direct workflows pass this gate in each main modality. MedGemma passes for CXR; its CT and MRI completion falls below the gate.

970 assignments: 480 main direct, 120 viewer, 270 role, 40 repeat and 60 fresh. There are no pending frozen assignments. Recovery-assisted results include 198 assignments with documented pre-dispatch setup failures.

[E1, E2] Qualified outcomes include properly structured abstentions. Setup and preprocessing are separate from inference latency.

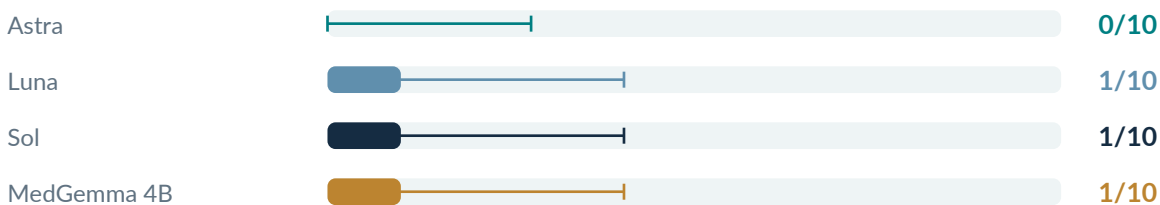
03 / FINDING RESOLUTION

Key report-positive findings were often unresolved.

Correctly resolved report-positive assertions / all explicit assigned positives. Failures and abstentions stay in the denominator. Whiskers: Wilson 95% intervals.

■ Astra ■ Luna ■ Sol ■ MedGemma 4B

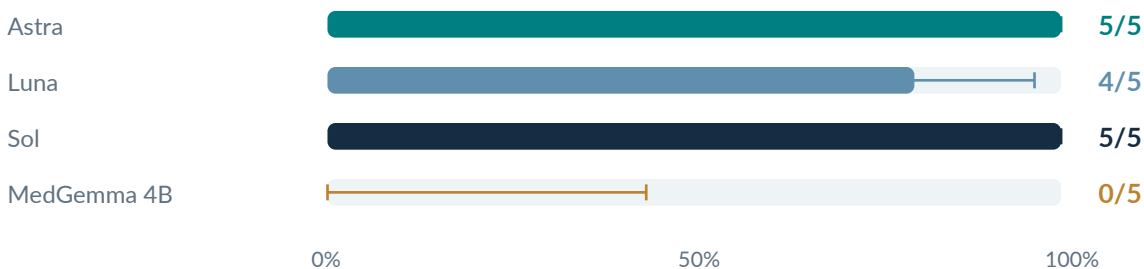
Pneumothorax / 10 positives



Pulmonary nodules / 19 positives



White-matter signal assertions / 5 coded positives



MRI is a candidate for narrower validation, not a validated ranking

Two of the five coded MRI positives have normal-impression conflicts. A post hoc source-scope restriction leaves GPT concordance at 3/3 and MedGemma at 0/3. The original 5-case scores remain unchanged. No explicit target negatives support MRI specificity.

[E1, E3] All 10 pneumothorax and 19 nodule-positive source mappings were independently audited. Report support does not prove visibility in supplied frames.

04 / SPECIALIST VALUE

More inference did not establish a general specialist advantage.

270 role assignments cover delegated reads, independent second-reading and self-review. The same saved MedGemma study read is reused; it is not a new physical GPU read each time.

Same-case controls: 65 of 66 estimable cells show zero difference

The sole nonzero cell is Luna/CXR cardiomegaly, second-reader versus self-review: +1 resolved assertion in 4 supported pairs (+25 percentage points; exploratory interval 0 to 75; exact McNemar p=1.0). Six additional cells have no reference support.

OBSERVED TRANSITIONS ACROSS SUPPORTED ASSERTIONS

Condition	Comparison count	Gained / lost resolution
CXR / delegated	195	1 / 2
CXR / second-reader	195	1 / 1
CT / delegated	42	0 / 1
CT / second-reader	42	0 / 0
MRI / delegated	12	0 / 2
MRI / second-reader	12	0 / 1

These are repeated finding/model comparisons, not independent patient counts. Resolution changes include movement into or out of unresolved states. The sole completed-label reversal occurs in a source-conflicted MRI case and is not adjudicated clinical harm.

Incremental complexity requires demonstrated value

On identical Sol controls, median component sums are CT: 54.0s delegated / 82.0s second-reader / 45.0s self-review; MRI: 106.2s / 125.7s / 50.3s. These omit setup/preprocessing and are not serial wall time or dollar costs.

[E1, E4, E5] Exploratory controls were added during audit. Sparse zero differences establish neither benefit nor equivalence.

05 / EVIDENCE BOUNDARIES

Input fidelity and reference quality limit what the results mean.

Observed failures belong to the tested workflow. They cannot be attributed to model weights alone, or generalized to full clinical examinations.

Material limitation	Decision impact
MedGemma montage distortion	CT montages: 2048 x 1080 to 896 x 896, about 1.90x aspect change. MRI: about 1.26x. Revisit evidence delivery before judging intrinsic volumetric ability.
GPT native image transformation	Audited MRI inputs: 2048 x 1620 to 1779 x 1408, approximately aspect-preserving. Delivered pixels differ from source attachment bytes.
Viewer contamination and sparse coverage	18 viewer reads had prior conversation context. Both sides of these pairs are excluded. A clean CT trace inspected 9 distinct slices out of 247 frames.
Existing reports, no new clinical review	Assertion concordance is measured. Translation, narrow coding, conflicts and unknown public training exposure limit interpretation. Error severity is unadjudicated.
Missing comparative evidence	Historical GPT controls, MedGemma 27B, broad modality coverage and limited differential diagnosis were not tested. No validated overall model ranking.

REPEATABILITY IS SEPARATE FROM CORRECTNESS

Identical finding-status objects: Astra 7/10 repeat pairs, Luna 7/10, Sol 6/10. MedGemma is identical in 8/8 pairs with two qualified responses; 2/10 pairs lack two qualified responses. Repeated agreement does not turn an incorrect finding into a correct one.

Fresh studies do not rescue the main diagnostic claim

Fresh CXR cardiomegaly: Sol 1/5, MedGemma 3/5. The sole CT nodule positive is unresolved by both. The narrow primary MRI white-matter endpoint has no coded fresh positives; known omitted clauses remain unscored. No retrospective relabeling was used.

[E1, E3, E6] Fresh completion: Sol 10/10 in each modality; MedGemma CXR 10/10, CT 6/10, MRI 7/10. These are execution measures.

06 / ACTION AGENDA

Make the next study narrower, cleaner and decision-specific.

Recommended sequence for future authorized research. These steps have not been executed as additional benchmark conditions.

Priority	Action	Acceptance evidence
1 / Retain	Keep isolated attachment and viewer engineering with explicit response and identity checks.	Trace audit passes; setup reliability is reported separately from response capture.
2 / Repair evidence delivery	Preserve montage geometry; inspect native delivered pixels and anatomical coverage.	Known source-to-model transformations; representative lesions fall within delivered evidence.
3 / Tighten existing-report references	Reconcile original-language clauses, translation and normal-impression conflicts. Use explicit positive and negative assertions.	Versioned reference rules; conflicts excluded prospectively; no live-doctor dependency introduced.
4 / Preregister one narrow endpoint	Choose a finding and input method; use institution-separated cases and a paired self-review control.	Prespecified denominators, missing-outcome handling, intervals and precision target.
5 / Decide specialist value	Add consultation only if matched controls demonstrate useful benefit with measured incremental resources.	Benefit supported beyond sparse observations; full workflow components accounted for.

Planning guidance, not a clinical safety guarantee

Approximately 385 independent positives and 385 negatives per task give a worst-case normal-approximation 95% margin near +/-5 percentage points. Zero misses in 59 independent positives gives a one-sided exact 95% upper miss-rate bound below 5%. Reference quality and independence still matter.

[E1, E7] Patient-care deployment remains on hold under the current evidence. Hardware, subscription-dollar allocation and whole-lifecycle ROI are unmeasured.

07 / ACCOUNTABILITY & SOURCES

Traceable results. Explicit accounting boundaries.

Evidence completed 1 October 2026. Original diagnostic outputs and frozen scores were preserved; no new inference was used to create this report.

Measured resource	Observed amount	Boundary
GPT diagnostic usage	66,584,015 input tokens 405,317 output tokens	Includes development and native-only receipts; excludes curation/orchestration. Cached input is a subset.
Local MedGemma	172 timed serving records 6,226.7 inference seconds	Includes development; 160 frozen physical study reads. Plus 19 pre-inference guards and 3 records without duration.
Account actions	One authorized free reset No purchases or upgrades	Shared compute and quota; no dollar/energy estimate or equivalence claim.
Conversation accounting	1,002 observed 970 frozen assignments	31 development + 970 frozen + 1 auxiliary. Original 1,000 cap corrected prospectively; no new study condition.

HOW TO INTERPRET THE NUMBERS

Qualified outcome: a terminal response meeting engineering validity checks, including structured abstention. **Finding resolution:** a correct explicit report-supported label under the assigned workflow; unresolved cases remain in the denominator. **Intervals:** Wilson 95% for fractions; paired comparisons are exploratory and not multiplicity-adjusted. Global patient independence is not fully certified.

EVIDENCE INDEX

- E1 Decision report and complete technical appendix
- E2 Execution metrics and final acceptance
- E3 Task-specific counts, intervals and missing-outcome bounds
- E4 Identical-control comparisons and specialist transitions
- E5 Role timing and component cost attribution
- E6 Independent source, arithmetic and native-boundary audit
- E7 Statistical methods, provenance and requirements audit

Source version: a6b126b. Full requirements and gaps: REQUIREMENT-AUDIT.md. The 90-artifact delivery-checksums.sha256 manifest verifies the retained package; this PDF is an executive synthesis. Protected reports and raw traces remain local.

CXR = chest radiograph. CT = computed tomography. MRI = magnetic resonance imaging. MedGemma here = pinned 1.5 4B NF4 workflow; Sol = gpt-6.1-sol.